

Royal Netherlands Institute for Sea Research

FAIR - UNECSS workshop

Best Practices for Making (Meta)Data FAIR

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What does a FAIR data package look like?

A FAIR data package is the data in open format:

- Csv files
- jpeg files
- Ect.



These 4 documents:

- 00_README.txt — the entry point: what this is, and what is in the folder
- LICENSE.txt — what are you allowed to do with it
- metadata.csv — the data dictionary: what every column in every file means
- provenance.txt — the origin story: who, what, where, when, how

and supporting files:

- provenance_files/ — protocols and calibration logs

When does each document happen?

During Planning phase:

- Data Management Plan
- Choose repository
- Choose File formats
- Metadata standard
- Controlled Vocabularies



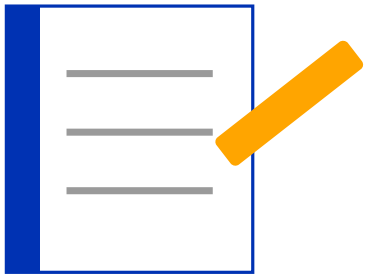
When does each document happen?

Day 1 of collecting data:

- Start with the README draft
- provenance.txt (who / what / where)

During research processing and collecting:

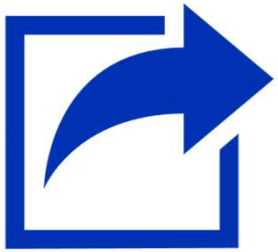
- Append every processing step to provenance.txt, as you do it
- Write metadata.csv as soon as the column headers stop changing



When does each document happen?

Before sharing:

- Add license
- Update metadata.csv
- Update README
- Check provenance
- Fix version and date



00_README.txt

Contents:

1. Dataset identification
    - Title, DOI, version, release date
    - Campaign / cruise, survey period
    - Geographic scope, bounding box
    - Keywords
    - Funder and grant

2. Licence and access
    - Licence + URL, embargo status
    - How to cite
    - Contact + ORCID

3. File inventory
    - Every file/folder, one line, what it holds

4. How the files relate
    - Table keys

5. Methods (short summary)

6. Controlled vocabularies used

7. Related resources
    - Qualified links, with PIDs

00_README.txt

Some general tips:

At the start:

Just copy info from you DMP

- Description, data types, people involved are all already written down
- It's okay if things are incomplete!

README grows with the project:

- It's like house keeping
- Keep version control (change log, git)
- Add the DOI, the citation, First release date, version 1.0 when publishing

Write it for someone who has never met you and never will

LICENSE.txt

In most cases…
Just copy paste or link via PID

The simpler the license, the better.

Yay open science!

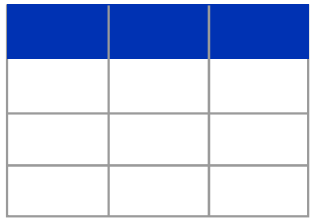
But keep in mind:
Creative Commons licenses can **not be revoked!**

metadata.csv

Data dictionary:

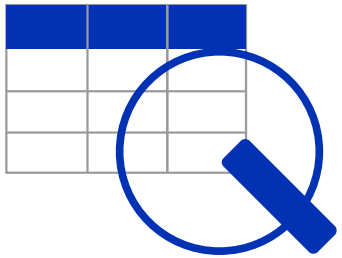
one row **per column** of **per data** file in the package:

- Column_name: Which column
- File_name: Name of file
- Definition: plain language, no acronyms, be verbose
- Vocabulary_uri: URI of the vocabulary used for the column
- Data_type: string, integer, decimal, datetime, etc.
- Unit: The unit of the column
- Unit_uri URI of the unit
- Example_value : one real value from the column (42)
- Missing_value_token: the **single** token used for missing, e.g. NA



metadata.csv — one row, in practice

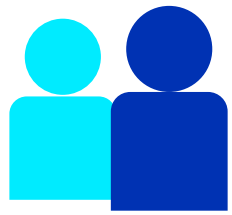
Column_name	File	Definition	Vocabulary_URI	Data_type	Unit	Unit_URI	Example_value	Missing_value_token
Sea_temperature	measurements.csv	in-situ sea water temperature	http://vocab.nerc.ac.uk/collection/P01/current/TEMPPR01/	Decimal	Degree Celcius	http://vocab.nerc.ac.uk/collection/P06/current/UPAA/	9.3	NA



provenance.txt

Who = everyone who contributed:

- Name, role, what they did, ORCID; institute with ROR
- the rights holder



What = the materials the data came from:

- Instruments with serial numbers, settings and calibration dates
- Protocols, as files in **provenance_files/**
- Software + versions, and the processing script
- The source data with a link or PID + link provenance source data

Where = the places involved:

- Sampling area with MRGID, Sample locations in decimal degrees
- Labs the samples are processed and stored
- Servers where the data is stored

provenance.txt

When = Add time stamps to:

- Samples collected
- Data created
- Data processed
- Dataset published
- Dataset updated

How = every processing step, in order:

1. Collected samples with CTD approximately 1 meter above the seabed floor.
2. Converted raw .hex files to readable csv format using calibration (see sbe43_calibration_log.txt)
3. Quality control Etc.

provenance.txt

What you don’t need to do:

You don’t need to write “why” in provenance.txt
It is not a How-to or manual

Don’t write what you plan to do, only what you have done.

- Incomplete is better then false

Better done than perfect

- Messy, ugly, unstructured, informal, incomplete provenance is better than no provenance

Quick summary

1. Before start: DMP
2. Day 1 of project: README skeleton + start provenance
3. Every processing step: append it to provenance
4. When headers final: write metadata.csv
5. Before sharing: add license, update metadata, fix version and date
6. At deposit repository: DOI and citation into the README then publish
7. Afterwards: keep changelog, update metadata when needed.
8. Keep metadata forever

Write for the unknown other.

